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World's Longest 9-5/8" Cased Hole Gravel Pack with Multizone Alternate Path System in Offshore Malaysia

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Abstract

PETRONAS completed well K on field KNB in offshore Malaysia by pushing the limit of the distance between two target perforation zone with 9-5/8" Cased Hole Gravel Pack Alternate Path System (APS). This Alternate Path System installation is the world's longest successful application of the technology where the two target zones are separated with 3,300ft of blanks and the Gravel Pack treatment is pumped in a single trip.

Multizone Alternate Path Gravel Pack System has been successfully installed in various field in the world, allowing multiple target zone to be treated at the same time through a transportation tube installed on the blank pipe and screens commonly known as shunt tube. This technique eliminates the need for multiple assembly run, perforation, wellbore clean up, and gravel pack pumping. Additional challenge arises in well K where the two target zones are 3,400 feet apart, and a trajectory with maximum deviation of 60 degrees, posing a risk of not being able to reach TD and perform the gravel pack treatment in an effective manner. The main gravel pack treatment was performed using viscoelastic surfactant (VES) fluid, which specifically designed to a certain viscosity allowing the fluid to flow through the shunts (high shear environment) while maintaining it's proppant suspension capability. Intensive discussion and simulations has been made, specifically for the tubing movement and feasibility checks for all interfacing components. Lessons learned from previous jobs are incorporated to ensure the operation runs smoothly.

Installation of the 9-5/8" Cased Hole Gravel Pack with Alternate Path System was successful. The completion string has 3,300 ft of blank pipe section, and swellable packer was used to isolate zones that can be of interest to be exploited in the future. The Quantum Packer was set, and a good annular pressure test was obtained. The tool was then released from the packer and then moved to reverse position to perform tubing pickle. The alternate path system features one main flow path from the tubing which allows the reverse friction test, circulation test, injectivity test, and matrix acidizing to be performed on the top zone only. Once the acid treatment of the top zone is done, a screen out drill is performed prior to the main gravel pack treatment in which screen out is achieved as planned, and the service tool was moved to reverse position

to clean the string from remaining gravel and pulled out of hole. The upper completion string was then run in the hole as per tally and landed successfully.

The success of the world's longest 9-5/8" Cased Hole Gravel Pack with Alternate Path System has shown that with excellent planning, proper risk assessment, and flawless communication, anything is possible. This success also highlights SLB's Alternate Path System's ability to significantly reduce operational time in multiple zone sand control installation and HSE exposure, which in turn makes a profitable and economical well.

Introduction

Gravel pack has been a tried and proven sand control method for years. It is done by using a combination of two types of filtering media: screen, and gravel. The gravel, or proppant, is pumped with conveying fluid through the work string and service tool. It is sized between 5 – 6 times the formation median grain size (Saucier, 1974). The screen size / pore opening should be fine enough to retain the formation sand/proppant, and coarse enough to minimize impairment. According to Penberthy and Cope, 1960, the screen slot / pore opening size should be between 50% to 75% of the smallest gravel size. As time goes, this basic method has been applied and modified to suit various well conditions: vertical, horizontal, cased hole, open-hole, single, or multiple zones. When done properly, the combination of screen and proppant pumped downhole through the service tool can effectively block sand formation from coming to the surface. The result is sand-free production throughout the well life.

Despite the superiority, this method also has its demerits. Introducing an abrasive material like proppant through a string of pipes that needs to move from one position to another means a high risk of having the service tool stuck. Another thing to consider is that it's still the highest costing compared to the other sand control method such as stand-alone screen, or chemical consolidation. This is due to the nature of the treatment itself which is combining hardware with gravel as the filter media. Technical complexity such as well angle, perforation / open hole interval, bottom hole pressure, temperature, frac pressure, and other reservoir conditions are the factors that can increase both the risk and the cost of a gravel pack treatment. However, further technology advancement has allowed the optimization of the technique despite the challenges. Multiple zones can now be treated in one run by single trip multizone technology, long open hole section can be treated using alpha-beta technique and combined with alternate path screen, more advanced carrier fluid allowing less friction while transporting proppant, are among the example of the new technology.

The case of well K, located in the Sabahan Waters of Malaysia, can be one of the examples of how to further optimize the basic gravel pack completion to allow economically viable operation using the technology that is currently available. Referring to [Figure 1](#), this is a gas well comprising five reservoir zones—two primary reservoirs and three future development targets – Behind Casing Opportunity (BCO). The well features a maximum deviation of 60 degrees and extends to a total measured depth of 13,745 feet (MDDF). The very long distance between the two main reservoir zones is the main challenge of this well.

The conventional approach to this type of well is to complete each zone separately. Which means that there are four separate runs that needs to be done in the lower completion section, which are: RIH sump packer, RIH and perform gravel pack for lower zone, RIH intermediate completion, and lastly RIH and perform gravel pack for upper zone. This option is not only costly but impose higher risk as multiple run-in hole means multiple risk of tool getting stuck, and longer working time for the personnel off-shore. The challenge is to ensure that the gravel pack hardware can be deployed safely, proppant can be carried through the long interval and successfully packing the casing-screen annulus to prevent formation sand from being produced while keeping the well economically optimized.

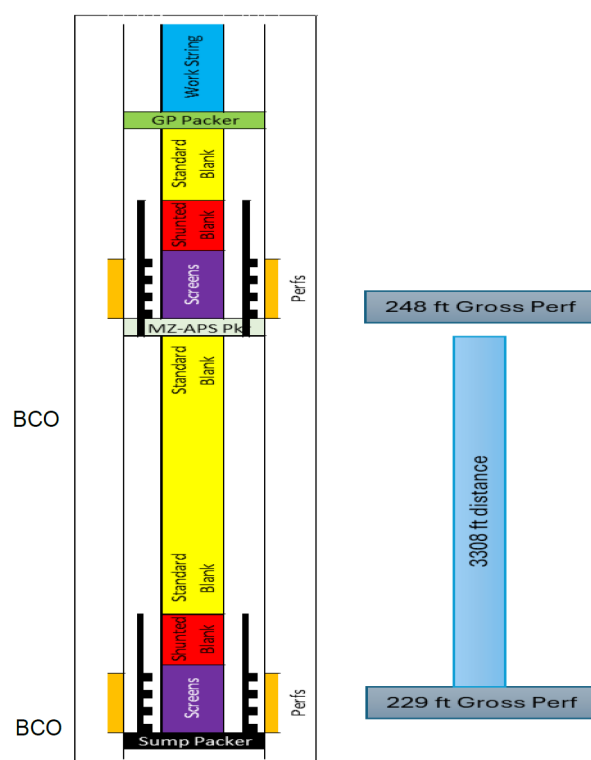


Figure 1—Well K Completion Illustration

Completion Design and Execution

The completion design of Well K consists of a dual-selective completion with dual isolation packer on the upper completion. To overcome the challenge of completing two different zones with very long interval, a 9-5/8" Single Trip Multizone Alternate Path System was utilized. It comprised of Alternate Path Screen, a screen with nozzle shunt tube attached using steel ring, it is a reliable path for the proppant slurry to reach any region in the wellbore. When bridges formed during pumping, the shunt tube allows slurry to by-pass it, ensuring complete pack regardless of downhole condition. The nozzles on the shunt tubes are spaced strategically to achieve full annular packing, which means that even the non-perforated interval can still be efficiently packed through the screen.

To isolate between the two reservoir zones to be produced, an MZ APS Packer is used. It is a cup-type packer with shunt tubes attached through the element. The cup-type element offers instant zonal isolation as no hydraulic pumping / mechanical manipulation is needed to set the packer, while the shunt tubes that runs through the elements provides flow-path of the slurry or any other fluids that is pumped during any of the completion stages. This feature is an important feat as it allows both of the reservoir zones to be completed and treated at the same time, cutting the run-in hole count by half to only 2 runs of RIH the sump packer, and RIH and performing gravel pack for both upper and lower zones.

Aside from the targeted reservoir zones to be produced, there are several other reservoirs that is planned to be accessed in the future. These Behind Casing Opportunity (BCO) Reservoirs needs to be accessible for the future interventions. With this in mind, the sump packer and blank pipes are strategically placed around the BCOs to allow future access. Another feature of the completion design is the combined use of both 5.5in and 3.5in blank pipe. The 3.5in blank pipe is used in the long section between the upper and lower zone. The use of smaller blank pipe allows the total cost of the completion to remain low as it is not as expensive as 5.5in blank pipe. Smaller blank pipe also allows better proppant slurry down-flow path during gravel pack treatment. A swellable packer is placed below the BCO zone and above treated zone. Once the swellable packer fully swells, it isolates both lower producing zones, and the BCO area.

A comprehensive PSD analysis using actual sand core sample was performed to provide better understanding of the sorting coefficient, uniformity, and fines content of the formation sand. Once the PSD result is obtained, a sand retention test (SRT) was done to determine the best combination of gravel and sand screen size for the formation. This step is paramount to ensuring optimal well productivity by maximizing permeability and longevity under typical production and treatment conditions. Adhering to the Saucier's method of sizing the gravel 5 – 6 times the median size of the formation sand, while the screen sizing is within 50% to 75% of the smallest gravel size, the following combination of screen and gravel size has been obtained as summarized in [Table 1](#):

Table 1—Gravel and Screen Size Selection

Gravel Sizing	20/40 mesh
Screen Sizing	Shunted APS Wire Wrap Screen, 12 gauge (150 micron)

With the given lower completion tool configuration, the proppant placement will be sequential, ie: Slurry will fill the upper zone until its fully pack, before its travel via the alternate path to fill in the Lower zone until the screen out achieved. As an extra friction is encountered when the shunt activated, a pump rate is lowered in order to maintain the pressure within the allowable value.

There are two option gravel pack carrier fluid, polymer and Viscoelastic Surfactant (VES) base. VES base fluid is known in the industry for having flexibility of mixing and adjusting properties on fly during pumping while providing lower friction as well as better retain permeability compared to its polymer base GP fluid alternatives, hence the VES base fluid is selected

[Figure 2](#) below illustrates the completion schematic of Well K with two 2 long zones interval with long unperforated casing.

The service tool utilized to convey the proppant slurry also plays essential role in completing the well. It comprised of a multiport sub below the setting module to allow slurry to be pumped at high rate. Below the multiport sub, a ball valve that opens and closes with a collet movement through a J shaped mandrel. The ball valve allows set down capability to provide weight and prevent tool movement during pumping at high rate. The ball valve provides a full-bore capacity for the fluid flow during pumping. With the challenging well trajectory, a Torque and Drag analysis need to be performed to ensure that the assembly can reach TD safely and that enough set down weight is available when pumping the gravel pack. Based on the T&D result, 50 joints of 4.75in Drill Collar needs to be used in the bottom section of the string to reach TD. Furthermore, an additional 40Kips set down weight is needed to ensure no tubing movement occurred during high-rate pumping.

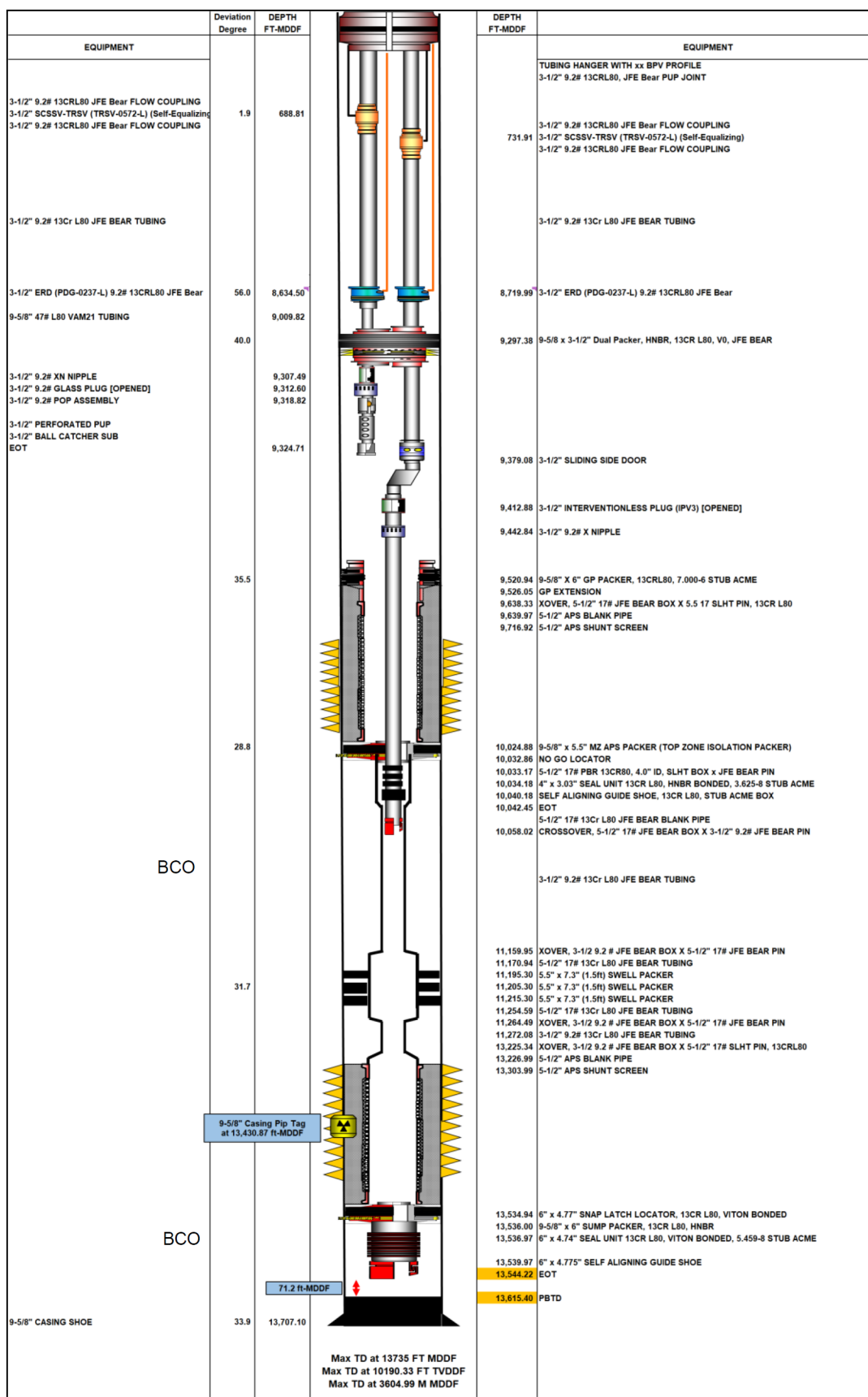


Figure 2—Completion Schematic of Well K

To get better understanding of the downhole event during pumping, downhole gauges are installed in both the drillpipe above service tool and also in the wash pipe behind the screen. The downhole gauge placement is illustrated in the [Figure 3](#) and summarized in the [Table 2](#) below. The data obtained from the downhole gauge is used for post job analysis.

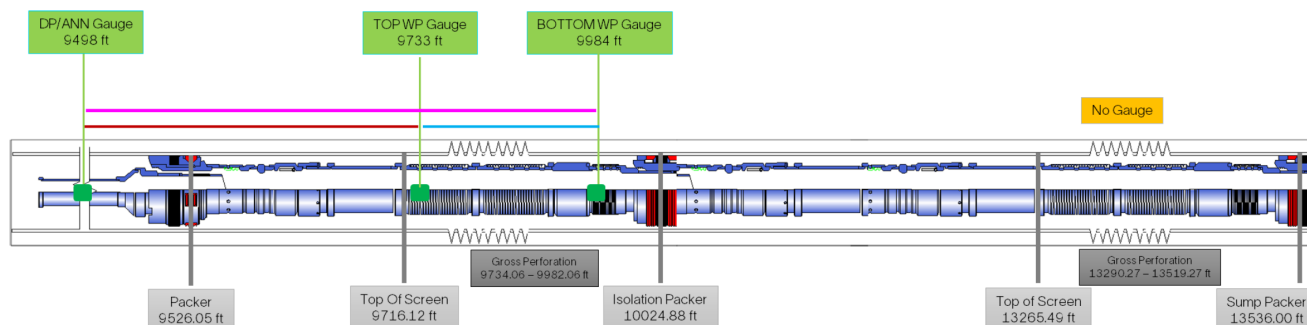


Figure 3—Downhole Gauge Placement

Table 2—Downhole Gauge Placement

DP – TOP	Section between DP internal-reading gauge to TOP Washpipe external-reading gauge
TOP – BOTTOM	Section between TOP to BOTTOM Washpipe external-reading gauge
BOTTOM - ANNULUS	Section between BOTTOM Washpipe to DP external-reading gauge

Gravel Pack Treatment

The performance of Gravel Pack (GP) operations is not solely reliant on hardware and service tools but is significantly influenced by the carrier fluid, proppant type, and pumping treatment employed. In an effort to enhance sand control performance, a viscoelastic surfactant (VES) base is utilized as the carrier fluid. VES offers various advantages such as rapid flowback and fracture cleanup, tolerance to varying water qualities, favorable fluid rheology, and efficient proppant transport. Compared to polymer-based fluids, the mixing process VES fluid is conducted on-fly using liquid additive pump, allowing for real-time viscosity adjustments which provide flexibility and reducing mixing time by up to four hours.

The GP pumping treatment for all zones followed a structured process as depicted in [Figure 4](#). The final pumping schedule was fine-tuned based on data analysis during circulation and step rate tests. Parameters such as extension pressure, Instantaneous Shut-In Pressure (ISIP), closure pressure, and fluid efficiency were defined through decline analysis conducted post step rate test.

To optimize execution efficiency, several offline rig activities were carried out, including rigging up and pressure testing the surface pumping workstring and pre-mixing acid for matrix acidizing steps to save online rig time.



Figure 4—Treatment Sequence

Tubing Pickle

Referring to [Figure 5](#), once the tool in Final Reverse Position, the pickle treatment was pumped to ensure clean workstring for operation.

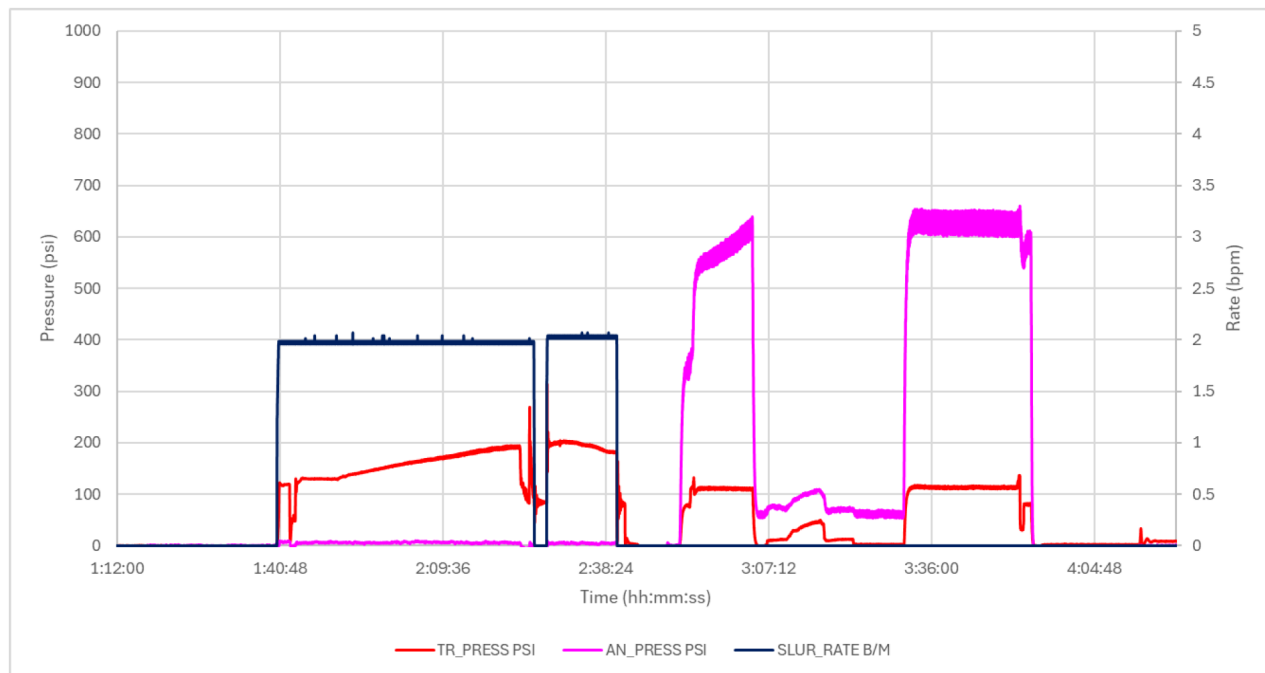


Figure 5—Tubing Pickle Chart

Friction Test

Figure 6 is the Friction Test Chart where the test was performed with the tool still in Reverse position to obtain data during pumping reverse out post Gravel Pack Treatment. Completion brine pumped by rig pump from annulus and return taken from the work string.

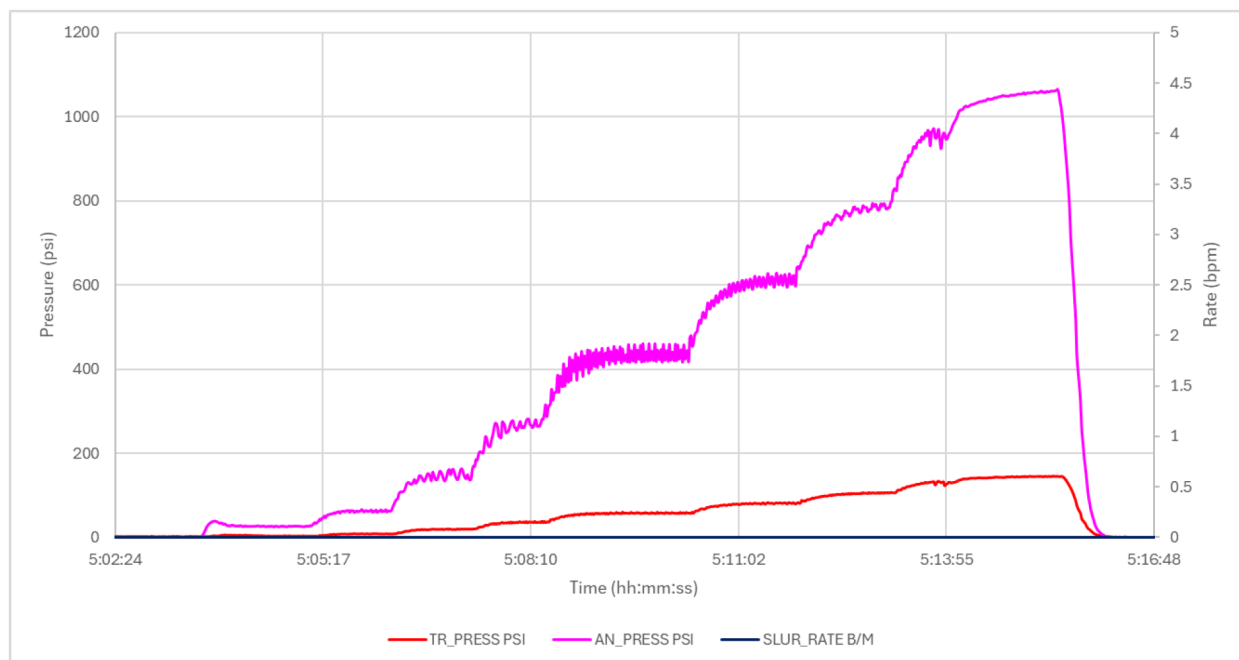


Figure 6—Friction Test Chart

Table 3 below summarizes the Friction Test Result for every rate.

Table 3—Friction Test Result

Rate (bpm)	Tubing Pressure (psi)	Annulus Pressure (psi)
1	3	25
2	8	63
3	19	148
4	36	261
5	57	427
6	81	601
7	106	793
8	143	1050

Pre-Acid Circulation Test

Referring to Figure 7, with Service Tool in Set Down (SD) Position, circulation Test was performed to measure dynamic fluid loss during pumping operation. In this test, the completion brine was pumped with SLB pump down the string and return was taken through the annulus side.

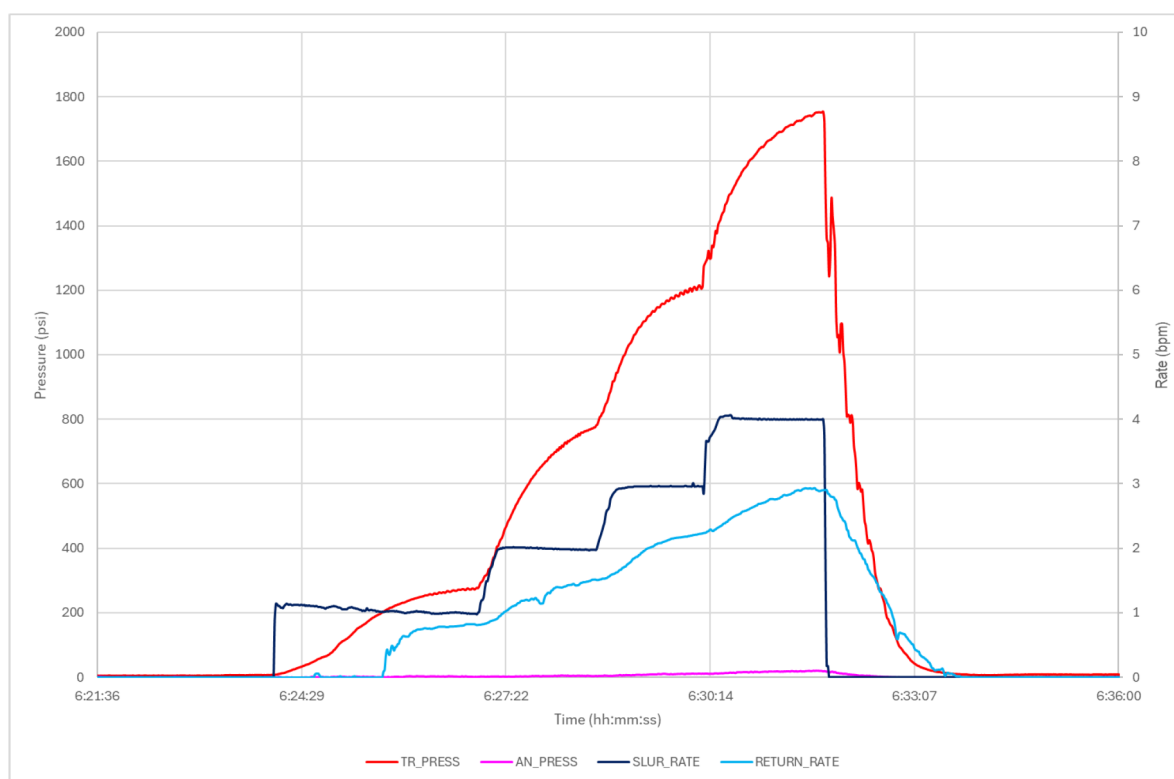


Figure 7—Pre-Acid Circulation Test Chart

Step Rate Injection Test

The next step (refer to Figure 8) was to perform injectivity test prior to pumping acid. The completion brine was pumped through the workstring while taking no returns and tool is located in Set Down (SD) Position. After breakdown was observed at 6bpm, the pump rate was gradually increased to up to 9 bpm. The result of this test will be the basis for performing the Matrix acidizing pump rate as well as becoming baseline for the Gravel Pack Job to be performed afterwards.

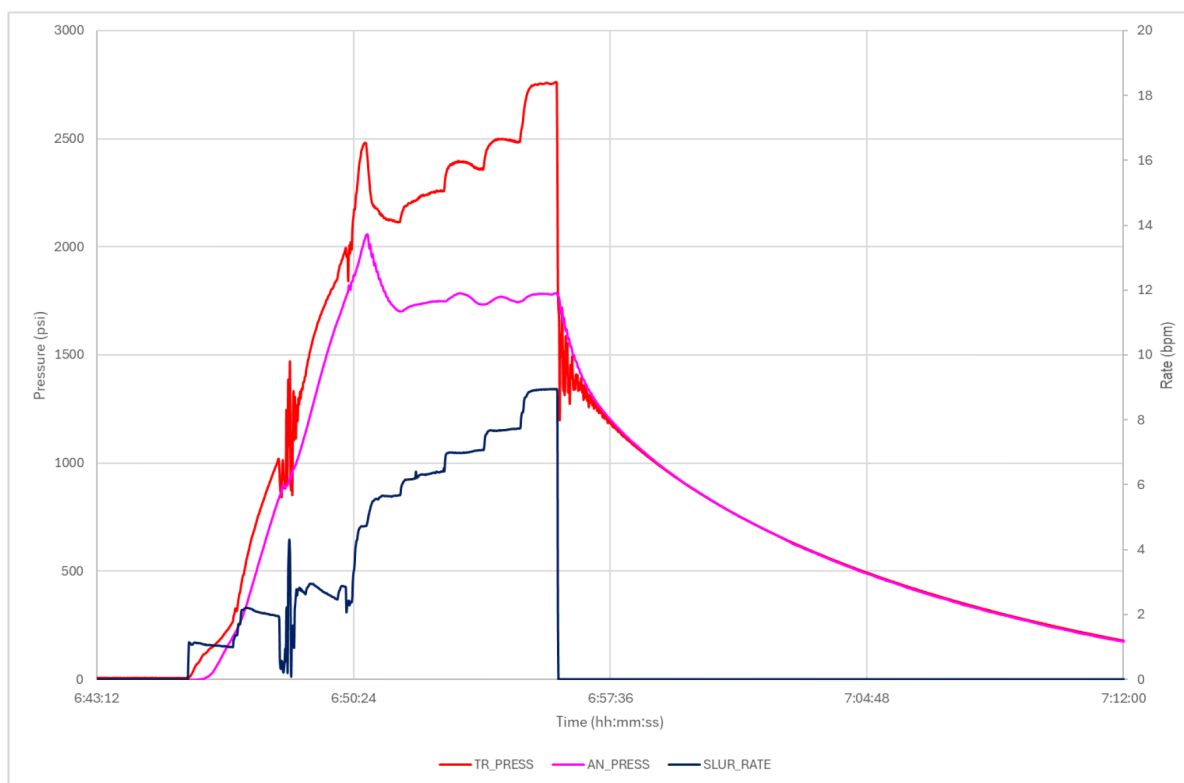


Figure 8—Step Rate Injectivity Test Chart

Step Rate Injectivity Test Analysis

Based on the Step Rate Injectivity Test Analysis as shown in Figure 9, the Extension Pressure = 4,805.22 psi and the Extension Rate = 4.44 bpm

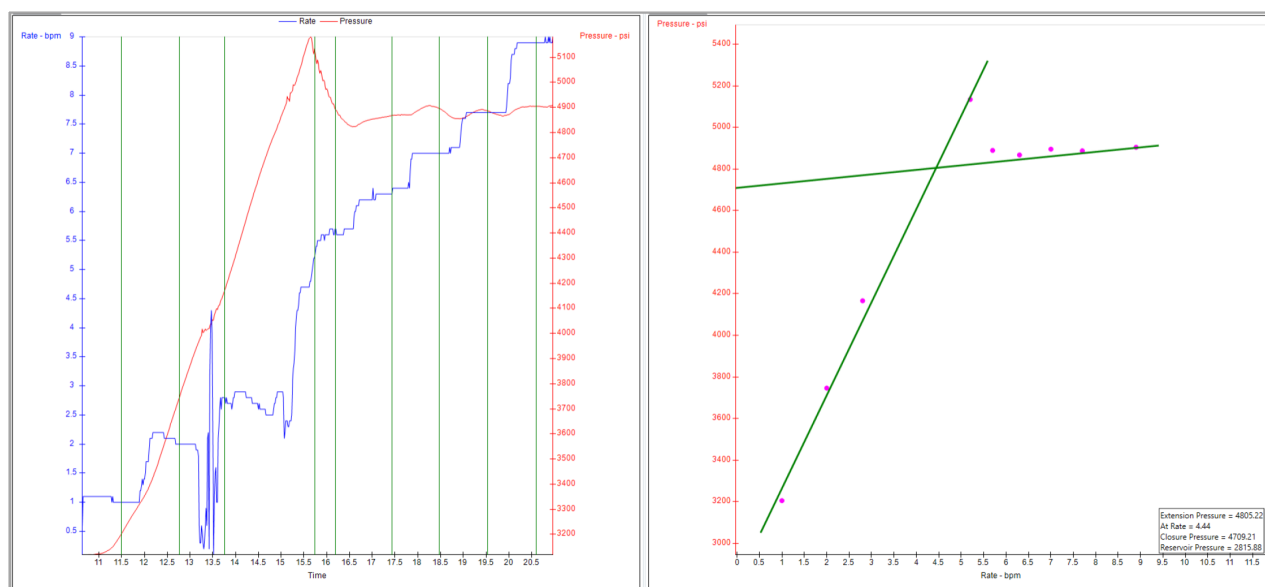


Figure 9—Pre-Acid Injectivity Test Analysis Chart

Matrix Acidizing

Referring to [Figure 10](#), matrix acidizing treatment fluid was split into 3 stages of Acid sequence with Diverter pumped in between the stages in order to distribute the fluids across the entire perforations. Mud Acid was used as main treatment fluid, while VES based fluid was used as diverter.

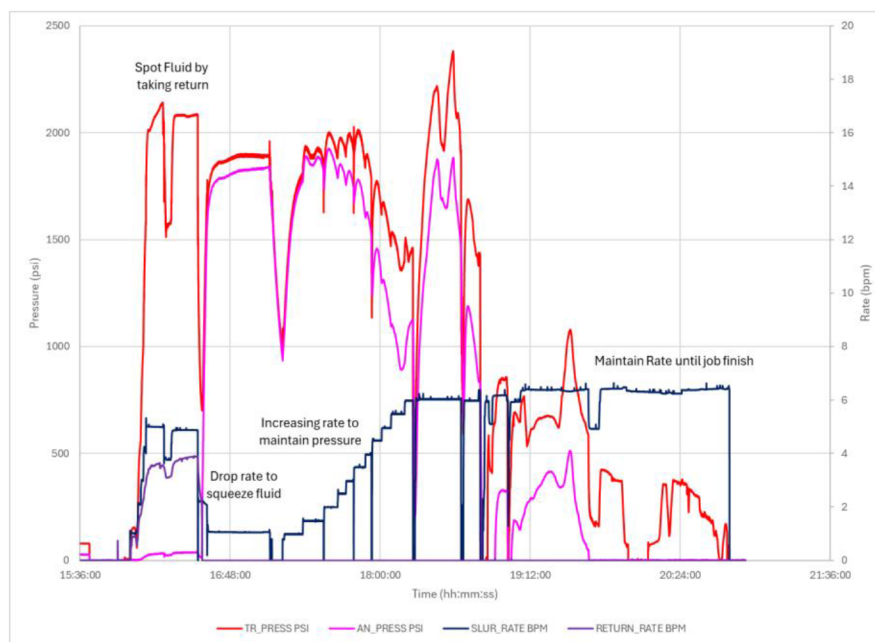


Figure 10—Matrix Acidizing Pump Chart

Treatment fluid was pumped with tool in Circulating Position (C) and SLB Choke fully opened to spot the fluids across the upper zone perforation before closing the choke to squeeze the treatment fluid into the formation.

Gravel Pack Treatment

The final pumping schedule was finalized based on data obtained during circulation, injectivity, and matrix acidizing step. The final pumping schedule for Gravel pack job is as follows.

Referring to [Figure 11](#), the treatment started by pumping slurry at high rate to fill more proppant inside the Upper zone. Once shunt activation pressure is observed, pump rate then gradually decreased to compensate for the high friction pressure while continuing pumping slurry to pack the lower zone (refer to [Figure 12](#)). Post job downhole data analysis shows the packing even during the treatment.

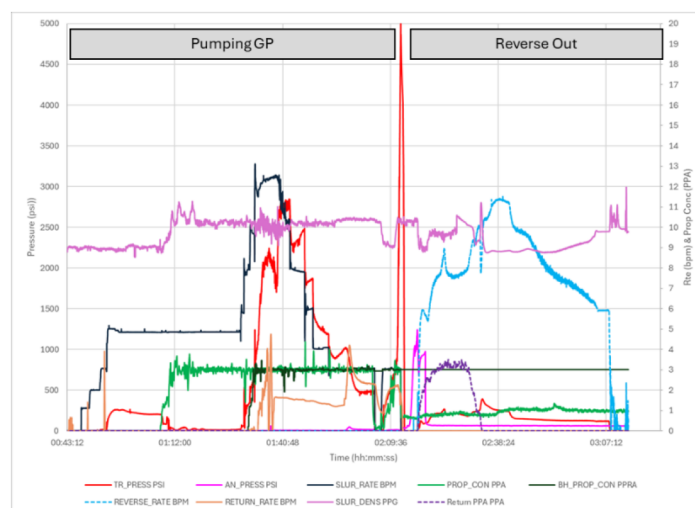


Figure 11—Pumping Job Overall Chart

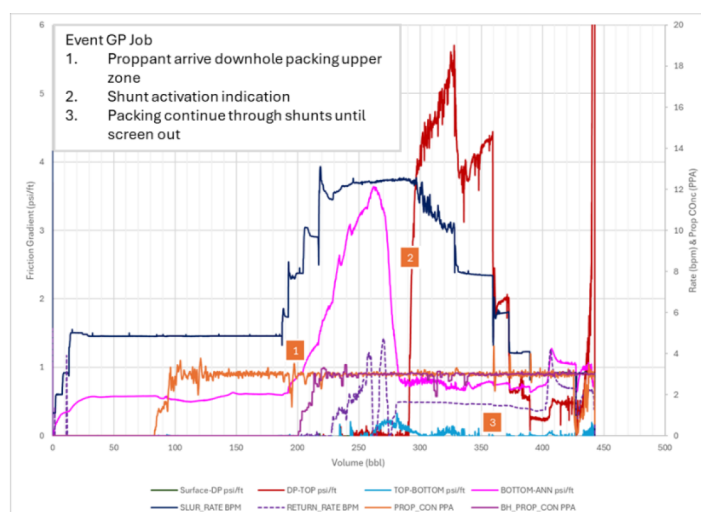


Figure 12—Pumping Job Sequence of Event Chart

Results and Discussion

Full annular packing achieved with good proppant placement within inside the top and bottom perforation intervals (refer to Figure 13). The summary of the treatment is as per below.

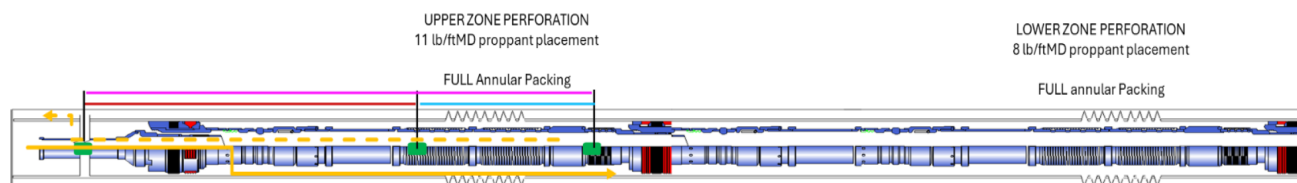


Figure 13—Treatment Summary

Conclusions and Lessons Learned

Conclusion

The MZ APS system has demonstrated its effectiveness in enabling multizone treatment as a single pumping stage, even when zones are separated by significant distances. This approach has resulted in notable time

savings compared to conventional systems. Additionally, full annular packing was successfully achieved during the pumping operation, confirming the system's reliability. The use of VES-based fluid further simplified operations due to its flexibility and capability for on-the-fly mixing, enhancing overall efficiency.

Lessons learnt

The operation highlighted the importance of conducting thorough torque and drag simulations, as well as tubing movement analysis, to ensure sufficient slack-down weight and prevent seal displacement during packer setting and pumping. Downhole gauge data proved valuable in verifying actual packing conditions both during and after the gravel pack job. Moreover, it became evident that designing sand control measures must be approached holistically—taking into account reservoir characteristics, execution strategy, and production goals. The perforation and fracturing strategy plays a critical role in influencing production outcomes, and an optimized design can significantly enhance well productivity, extend well life, and improve economic returns through increased recovery factors.

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Nomenclature

APS	: Alternate Path System
BCO	: Behind Casing Opportunity
ISIP	: Instantaneous Shut-In Pressure
GP	: Gravel Pack
MZ	: Multi Zone
PSD	: Particle Size Distribution
RIH	: Run in hole
SD	: Set Down
SRT	: Sand Retention Test
TD	: Total Depth
T&D	: Torque & Drag
VES	: Viscoelastic Surfactant

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